Shang Gao

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EDUCATION

Stanford University, Stanford, CA — Sep 2025 – Expected Jun 2027

- Master of Science, **Computation & Mechanics** (in progress), Civil & Environmental Engineering
- GPA: **3.986**/4.00
- **Official Research Assistant Appointments**
  Energy Control Lab - Prof. Simona Onori (Energy Science & Engineering) — Jun 2026 – Sep 2026
  Sustainable Systems Lab - Prof. Michael Lepech (Civil & Environmental Engineering) — Oct 2025 – Jun 2026

Tongji University, Shanghai, CHN — Aug 2021 – Jun 2025

- Bachelor of Science in **Civil Engineering**, Minor in Finance
- GPA: **4.50**/5.00
- Honors and Awards: National Scholarship (Top 2%, 2022–2023), Third-Class Scholarship (2021–2022), Undergraduate Social Activities Scholarship (2021–2023)

University of California, Berkeley, Berkeley, CA — Aug 2024 – May 2025

- Visiting Student, Tongji University-UC Berkeley 3+1 Program (Selected among 3 students from 400 undergraduates)
- GPA: **4.0**/4.0

PUBLICATIONS

- **Shang Gao**, Genghan Zhang, Mengjie Xie, and Kunle Olukotun. "HeuriGuide: Calibrating LLM Authority for Test-Time Heuristic Design." *International Conference on Learning Representations (ICLR 2027).* Under Review.
- **Shang Gao** and Abel Sanchez. "When Should Practice Stop? Evaluating Mastery Rules in Censored Platform Logs." *International Conference on Learning Analytics & Knowledge (LAK 2027).* Under Review.
- **Shang Gao** and Abel Sanchez. "Checkpoint Selection and Reward Comparisons in Offline Adaptive Practice." *ACM/SIGAPP Symposium on Applied Computing (SAC 2027), Artificial Intelligence for Education Track.* Under Review.
- Linjun Lu, Yuandong Pan, Jiechao Gao, **Shang Gao**, Varun Kumar Reja, Jie Wang, Michael D. Lepech, and Ioannis Brilakis. "AutoRoadAssess: A Multi-Agent Framework with Neuro-Symbolic Guideline Reasoning for Autonomous Pavement Condition Assessment from Multimodal Data." *Advanced Engineering Informatics.* Under Review.
- **Shang Gao**, Simone Fasolato, and Simona Onori. "A Low-Rank Multi-Output Gaussian Process for Cross-Cell Capacity Forecasting from Sparse Reference Performance Tests." *IEEE Transactions on Industrial Informatics*. In Preparation.
- **Shang Gao**, Yousef Liang, and Marco Pavone. "Temporal A2 Navigation Under Execution Changes." *Sim-to-Real-to-Field: The Science of Transfer (Sim2Real2Field), CoRL 2026 Workshop*. In Preparation.
- Lina Chi, **Shang Gao**, Huiyuan Xue, and Jie Wang. "Integrating Machine Learning and Domain Knowledge for Predicting Polyethersulfone Hollow Fiber Membrane Performance: A Comparative Study of CatBoost, XGBoost, and TabNet." *International Congress on Membranes and Membrane Processes (ICOM 2026).*
- Xiao Huang, Rossi Shao, Yue Xiang, **Shang Gao**, and Dipti Srinivasan. "Novel Physics-Informed Day-Ahead Wind Power Forecasting Model Design." *IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC 2026).*
- Jatin Aggarwal, **Shang Gao**, Amelie Bonde, Zhili He, Julia Gersey, Tomas Samuel Fernandez, Rahul Rejeev, Pei Zhang, and Hae Young Noh. "NeighborDrive: Privacy-Preserving Neighborhood Sensing Through Outreach Vehicles." *4th International Workshop on Social Infrastructure Systems (SocialSys 2026), ACM Sustainability Week Companion '26.* **Best Paper Award (2nd Place).**
- **Shang Gao**, Mingliang Zhou, Dayu Apoji, and Kenichi Soga. "Dynamic Training Strategies for Data-Driven Prediction of EPB Shield Tunneling Parameters: Balancing Accuracy and Computational Efficiency." *6th International Conference on Information Technology in Geo-Engineering (ICITG 2026).*
- Yiteng Lin, Qingtian Su, Fawas O. Matanmi, Xingfei Yan, and **Shang Gao**. "Mechanical Behaviour and Parametric Analysis of the Hybrid Girder Bridge Joint Between Steel–Concrete Composite Girder and Prestressed Concrete Girder." *Applied Sciences*, 2026.
- Yong Yuan, **Shang Gao**, Jiaolong Zhang, Xueming Yao, and Xinhang Xiong. "Earthworm-Inspired Bionic Robot for Geological Condition Exploration." Chinese Invention Patent CN116165090B, granted 2025.
- Bin Yang, Hongru Xiao, **Shang Gao**, Ke Lei, Wenshuo Chen, Qilin Zhang, and Congjun Wang. "Co-Experts-Based Large Language Model Question-Answering System for Building Construction." *Journal of Tongji University (Natural Science),* 2025.
- Xingyu Chen, Muran Yu, **Shang Gao**, Ying Liu, and Wei Shang. "From Viewing to Purchasing: The Moderating Effects of Lucky Draw in Live-Streaming Commerce." *Proceedings of the IEEE International Conference on Big Data (BigData 2024).*

RESEARCH EXPERIENCE

Physics-Based Modeling and Scientific Machine Learning

Finite Element Modeling and Machine Learning Acceleration for Reservoir Subsidence — Jun 2026 – Present

Supervisor: Prof. Ronaldo Borja (Civil & Environmental Engineering, Stanford University)

- Developed coupled finite-element models and machine learning surrogates to investigate long-term reservoir subsidence induced by hydrocarbon extraction and enable efficient multi-parameter prediction and inversion.
- Generated 1,000 parameterized axisymmetric FEM cases spanning 28 years using MOOSE/C++ and designed a hierarchical surrogate framework combining DeepONet with POD-reduced lightweight experts and a physics-input router.
- Achieved 2.33× faster full-field inference than standalone DeepONet at comparable RMSE (2.89 vs. 2.88 mm); integrated the surrogate into Bayesian inversion for uncertainty quantification and parameter identifiability.
- Extending the physical model to a 3D seven-well dual-porosity system to analyze pressure redistribution and persistent subsidence,

while developing training datasets to adapt the surrogate framework for multi-well spatial responses.

Physics-Informed Transformer for High-Temperature Gas Species Identification Jun 2026 – Present

Supervisor: Prof. Ronald Hanson (Mechanical Engineering, Stanford University)

- Developed a framework integrating physics-based spectral modeling with transformer-based learning for spectral line localization and atomic species identification from hypersonic shock-tunnel spectra.
- Compiled atomic line data from NIST ASD, Kurucz, and Meléndez-Barbuy (MB99), and built a forward absorption model using Boltzmann distribution, partition functions, and Voigt profiles; applied BH-FDR-controlled spectral matching to identify Fe, Cr, and Si.
- Inferred time-varying excitation temperatures and relative species concentrations from matched spectral lines using multi-line intensity ratios, and calibrated an empirical line-strength reference for 31 Fe/Cr transitions.
- Designed a Line-Set Transformer that integrates multi-scale convolutions, intra- and inter-species attention, theoretical spectral priors, and differentiable pseudo-Voigt reconstruction for joint spectral line localization and atomic species identification, achieving 96.9% line-detection F1 and 97.7% species-identification micro-F1 on an independent test set, while generalizing to real spectra with 92.5% recall and 87.1% F1.

Time-Dependent Constitutive Neural Modeling of Beef Mechanics Jan 2026 – Apr 2026

Supervisor: Prof. Ellen Kuhl (Mechanical Engineering, Stanford University)

ME 233 Course Project, Stanford University

- Developed constitutive artificial neural network (CANN) models to characterize the time-dependent mechanical evolution of Sirloin and New York Strip under refrigerated storage at 4°C (0, 1, and 2 days) using uniaxial compression data.
- Extended strain-invariant CANNs with logarithmic shear softening and independent exponential time-decay factors, using automatic differentiation to compute stress and identifying nonlinear constitutive parameters and storage-time coefficients.
- Achieved R^2 of 0.946–0.995 after L1-regularized constitutive term selection, revealing multi-rate energy decay in Sirloin but negligible temporal decay in New York Strip.

Physics-Informed Day-Ahead Wind Power Forecasting Oct 2025 – Mar 2026

Supervisor: Prof. Dipti Srinivasan (Electrical & Computer Engineering, National University of Singapore)

- Co-developed a physics-informed framework for 24-hour-ahead wind power forecasting that models residual deviations from turbine power curves to improve multi-step prediction.
- Enhanced the Sonnet frequency-domain forecasting architecture with learnable wavelet transforms, spectral-coherence attention, and a controlled Koopman layer incorporating yaw and pitch signals to capture multi-scale temporal patterns and turbine control effects.
- Incorporated rated power, ramp rate, Betz limit, and power-curve constraints during training and inference, achieving 35.33 kW MAE, 77.37 kW RMSE, and $R^2 = 0.9677$, reducing MAE by 9.1% versus iTransformer and 22.7% versus Crossformer.

Learning, Optimization and Computing Systems

LLM-Based Automatic Heuristic Design with Test-Time Inner Agents Mar 2026 – Present

Supervisor: Prof. Kunle Olukotun (Electrical Engineering, Stanford University)

- Proposed HeuriGuide, an LLM-based automatic heuristic design framework that introduces an inner LLM agent into test-time optimization and systematically studies how its decision authority should be calibrated.
- Designed three inner-agent organizations with progressively increasing procedural guidance (Solution Generator directly generates solutions, Strategy Adapter selects and adapts existing strategies, and Strategy Router selects only from curated strategies) to study the trade-off between agent autonomy, solution quality, and reliability.
- Evaluated HeuriGuide on the HeuriGym benchmark across nine combinatorial optimization problems in four domains, with Strategy Router increasing average QCI from 0.728 to 0.805 (+10.6%) over the outer-agent-only baseline and outperforming official baselines on 7/9 tasks.
- On E-graph Extraction, where the official HeuriGym baseline failed (QCI = 0), Solution Generator achieved 0.828 using the same GPT-5.4 model, demonstrating that test-time generation can solve tasks when curated strategies are insufficient.

Memory-Constrained Computation Graph Scheduling for ML Systems Feb 2026 – May 2026

Supervisor: Professor Kunle Olukotun (Electrical Engineering, Stanford University)

MLSys 2026 Competition Track A | https://github.com/yarongmu-google/MLSys

- Developed a graph scheduling solver for memory-constrained ML accelerators that jointly optimizes execution order, operator fusion, tiling, and tensor residency to reduce estimated computation and memory-access latency.
- Designed operator-fusion search seeded by MatMul along single-consumer chains; generated candidates with varying fusion thresholds and chain-length limits, while controlling search complexity via fingerprint deduplication, candidate pruning, and adjacent subgraph merging.
- Built a cost model and memory-aware scheduling strategy using tensor-value scoring based on computation cost, data reuse, dependency structure, and memory pressure to select optimized execution plans.

Evaluating Decision Rules and Offline Reinforcement Learning Policies for Adaptive Practice Jun 2026 – Present

Supervisor: Dr. Abel Sanchez, Research Scientist (MIT Professional Education)

- Studied adaptive mathematics practice using large-scale ASSISTments learning logs, examining how mastery rules affect practice termination and how checkpoint selection affects reward comparisons in offline reinforcement learning.
- Designed an auditing framework comparing Bayesian Knowledge Tracing (BKT), Exponential Moving Average (EMA), consecutive-correct, and sliding-window mastery rules using delayed future performance, coverage matching, common-trajectory comparisons, and two-way clustered confidence intervals by learner and skill.
- Found that coverage-matched BKT reached mastery 2.55 problems earlier than alternative rules on common trajectories; further

analyses showed that the BKT-consecutive-correct gap decreased from 1.73 to 0.94 problems under common-sample comparison, indicating that rule-specific sample selection had inflated the original difference.
- Trained one-step policy improvement and Implicit Q-Learning (IQL) policies under completion and predicted delayed-correctness rewards, with and without learner-history features, and extended the analysis to FQI, EMaQ, and CQL; compared checkpoint-selection rules using paired bootstrap confidence intervals and reward-permutation controls.
- Compared TD-loss early stopping, common training checkpoints, and validation doubly robust (DR) value selection; found that TD-loss checkpoint selection produced a 6.22-percentage-point difference between rewards in the probability of choosing the hardest problems, while the difference fell below one percentage point at a common training stage; DR-based selection removed the large contrast and improved estimated return by 0.86 percentage points.

Agent-Based Supply Chain Coordination for Emergency Response Oct 2025 – Jun 2026

Supervisor: Prof. Michael Lepech (Civil & Environmental Engineering, Stanford University)
- Developed an agent-based coordination system for disruption-aware supply chain operations, enabling rapid replanning under warehouse shutdown, road closure, and demand surge scenarios.
- Implemented routing and resource allocation optimization using OR-Tools with Adaptive Large Neighborhood Search (ALNS) to generate resilient delivery strategies under dynamic constraints.
- Built scenario-driven simulations integrating real-time traffic data (Google Routes API) to evaluate operational KPIs including service rate, delay, and energy usage.
- Designed a geospatial decision-support dashboard for visualizing candidate plans and disruption scenarios during emergency response.

Robotics and Embodied Systems

Temporal Policy Learning for Sim-to-Real Quadruped Navigation Jun 2026 – Present

Supervisor: Prof. Marco Pavone (Aeronautics & Astronautics, Stanford University)
Part of the TIAMAT Collaborative Research Initiative | https://tiamat.berkeley.edu/
- Conducted reinforcement learning research on quadrupedal locomotion and navigation, with a focus on transferring simulation-trained policies to real robots under complex terrain and observation delays.
- Migrated navigation tasks from Isaac Lab/Isaac Sim to mjLab/MuJoCo Warp and adapted multiple quadruped platforms, implementing a hierarchical control architecture in which high-level navigation policies generate planar velocity commands for low-level locomotion control; co-designed temporal policy architectures, navigation curricula, and reward functions for training across diverse terrains and obstacle scenarios.
- Evaluated feedforward, SSM, and causal Transformer navigation policies in an A2 case study under a unified training framework; across three random seeds and 27,648 simulation episodes, the Transformer achieved 79.2% average success and 0.438 SPL, with an 8.4% higher success rate and 19% higher path efficiency than the feedforward baseline.
- Studied what a navigation policy should retain about its recent interaction with the locomotion controller below it; two-second command or base-velocity histories raised mean success on A2 from 80.6% to 88.9%, while the latest command alone did not.
- Diagnosed an asymmetric cross-robot transfer failure in which Go2-trained policies failed on A2 across all three architectures; probing the observations with the robot state held fixed showed that the body-referenced terrain map lost the floor on the taller robot, and correcting that reference with all weights frozen recovered all nine policies.

Earthworm-Inspired Deep Earth Exploration Robot Sep 2022 – Sep 2025

Shanghai College Students "Creation Cup" Competition, Team Leader | Chinese Invention Patent (CN116165090B, 2025)
Supervisors: Prof. Yong Yuan & Prof. Jiaolong Zhang (Civil Engineering, Tongji University)
- Led the development of a bionic deep-earth exploration robot inspired by earthworm locomotion and EPB shield tunneling principles, creating designs using AutoCAD and conducting advanced 3D modeling using SolidWorks.
- Developed a Raspberry Pi-based control system and a power–torque analysis tool to optimize robotic mobility across heterogeneous soil conditions.
- Integrated multimodal sensing with embedded MEMS pressure, temperature, and humidity sensors to capture real-time penetration resistance profiles during soil penetration experiments.
- Built and tested a first-generation prototype, conducted soil penetration experiments and failure mode analysis, and iteratively refined the second-generation robot design.
- Won 1st Prize in the Municipal-level Competition and advanced to the semifinal round of the Global Innovation and Entrepreneurship Competition (Embodied Intelligent Robotics Special Track).

Perception, Vision and Infrastructure Monitoring

Object-Centric 4D Dynamic Scene Reconstruction Apr 2026 – Present

Supervisor: Prof. Iro Armeni (Civil & Environmental Engineering, Stanford University)
- Investigated object-centric 4D dynamic scene understanding and occlusion modeling, focusing on how to maintain object identity and motion consistency through partial and full occlusions and subsequent reappearance.
- Built synthetic datasets with Kubric and reproduced WorldTree, 4DPM, and NoPo4D to benchmark existing reconstruction methods, comparing scene feature extraction, spatial representation, motion modeling, and temporal processing under object motion and occlusion, and exploring directions for a new reconstruction algorithm.

Privacy-Preserving Vehicle Perception for Neighborhood Sensing Nov 2025 – Present

Supervisor: Prof. Hae Young Noh (Civil & Environmental Engineering, Stanford University)
Part of the NeighborDrive Project | https://web.stanford.edu/group/noh-lab/neighbordrive/
- Contributed to a multimodal mobile sensing platform integrating RGB, thermal, environmental, and GPS sensing for neighborhood-scale monitoring through outreach vehicles.

- Designed a vehicle detection approach that converts RGB frames into event images through temporal frame differencing to reduce exposure of privacy-sensitive visual information; fine-tuned YOLOv8s using RGB-generated pseudo-labels, achieving 0.79 precision, 0.69 recall, and 0.73 F1 on event images, over 10× the F1 of the unfine-tuned baseline.
- Analyzed 190,881 frames from nine driving videos to quantify the effects of event-generation thresholds and vehicle speed, establishing a recommended configuration with a macro-F1 of 0.658.

Automated Vision-Based Production Process Modeling for Cast-in-Place Construction Mar 2026 – Present
Supervisor: Prof. Martin Fischer (Civil & Environmental Engineering, Stanford University)

- Developed a vision-based framework to automate production data collection from construction-site video for construction process modeling and production control.
- Fine-tuned YOLO on nine months of construction-site video to detect and track workers and equipment, and constructed interaction graphs representing worker-equipment relationships.
- Trained GNNs to distinguish productive and idle periods in construction activities such as slab pouring and rebar wall installation, working toward automated extraction of activity timing, crew size, and idle duration.

Multi-Agent Pavement Condition Assessment with Neuro-Symbolic Guideline Reasoning Oct 2025 – Jun 2026
Supervisor: Prof. Michael Lepech (Civil & Environmental Engineering, Stanford University)

- Contributed to AutoRoadAssess, a neuro-symbolic multi-agent framework for automated pavement condition assessment that converts UK CS230 maintenance guidelines into executable knowledge graphs using local LLMs and a rule engine, enabling transparent defect grading, pavement condition rating, maintenance report generation, and one-time guideline parsing for efficient rule-based reasoning.
- Developed lane-level point-cloud processing using DBSCAN, RANSAC, and spatial overlay to assign pavement defects to lanes, preventing double counting in overlapping regions and supporting interactive visualization through Streamlit.
- Constructed and evaluated a 198-case benchmark for guideline reasoning, improving overall performance from 0.49 to 0.66 and cross-region reasoning from 0.48 to 0.81 over direct prompting.

Energy, Water and Climate Systems

Electrochemical Modeling and Capacity Prognostics for $LiFePO_4$ Batteries Mar 2026 – Present
Supervisor: Prof. Simona Onori (Energy Science & Engineering, Stanford University)

- Investigated $LiFePO_4$ battery degradation and capacity prognostics under constant-power cycling using physics-based and data-driven approaches.
- Extended a core-shell enhanced single-particle model to support constant-power cycling and identified structural and kinetic electrochemical parameters through multi-objective fitting of voltage, differential-voltage, and capacity responses.
- Built a physics-based capacity forecasting method by extrapolating aging-dependent parameter trajectories and feeding them into the electrochemical model for forward simulation.
- Developed a low-rank multi-output Gaussian process for cross-cell capacity forecasting, learning degradation patterns and cross-rate covariance from peer cells; compared it with physics-based forecasting across 16 cells under five aging conditions using RPTs 1–3 to predict RPTs 4–7.

Drought and Extreme Cold Effects on U.S. Thermal Power Generation Jun 2026 – Present
Supervisor: Prof. Ram Rajagopal (Civil & Environmental Engineering, Stanford University)

- Built a unit-day panel of all coal and gas units in the contiguous United States from CAMPD (2015–2023) with plant-level weather and balancing-authority demand, and compared each unit with its own non-extreme days; added a regional event design as well, because the daily comparison absorbs the season-long system response into its own baseline.
- Found that neither fuel lost generation on drought days on average, and that the losses appeared only in the driest tail, at a few coal plants that also went offline more often than in the same weeks of non-drought years; Western gas units instead ran about 23 TWh more during regional drought events, through more startups and higher loading rather than higher demand or heat.
- Traced the cold-weather risk to a few hours rather than to cold days: below $-10\,^{\circ}C$ both fuels generated more than on locally normal days, and with demand held fixed the gas increase was all demand while coal still ran about 9% higher; in the load-shed hours of Winter Storms Uri and Elliott, units that had run in the previous 30 days delivered about 60% (gas) and 73% (coal) of their measured winter capability, some 40 and 12 GW short.

Machine Learning for Membrane Performance Prediction toward Inverse Design Oct 2025 – Jun 2026
Supervisor: Prof. Michael Lepech (Civil & Environmental Engineering, Stanford University)

- Addressed the long experimental cycle and small-data challenge in PES hollow-fiber membrane development by building machine learning models that predict membrane performance from formulation and fabrication parameters to support formulation screening and process optimization.
- Built and compared CatBoost, XGBoost, and TabNet models using formulation, process, and structural features to predict flux, BSA rejection rate, tensile breaking force, and elongation at break; CatBoost achieved test-set R^2 of 0.954 for tensile breaking force and 0.865 for flux while improving elongation prediction from $R^2 = 0.401$ to 0.644 over XGBoost.

Geomechanics and Structural Engineering

Dynamic Training Strategies for EPB Shield Tunneling Prediction Sep 2024 – Jan 2026
Supervisors: Prof. Mingliang Zhou (Civil Engineering, Tongji University) & Prof. Kenichi Soga (Civil & Environmental Engineering, UC Berkeley)

- Investigated dynamic training strategies for machine learning prediction of operational parameters in Earth Pressure Balance (EPB) tunnel boring machines (TBMs), adapting model training to evolving conditions as excavation progresses.
- Evaluated Random Forest, XGBoost, and LSTM under sequentially updated Ring-by-Ring and Sliding Window training strategies for predicting penetration rate, cutterhead torque, and excavation chamber pressure.

- Developed a composite scoring framework integrating predictive accuracy and computational efficiency to compare models and training strategies, demonstrating that Ring-by-Ring achieved the highest predictive accuracy (R^2 up to 0.916 for penetration rate), while Sliding Window reduced training time by 37–42×.

Finite Element Analysis of Hybrid Girder Bridge Joints Aug 2025 – May 2026

Supervisor: Prof. Qingtian Su (Civil Engineering, Tongji University)

- Investigated the load-transfer mechanisms and mechanical behavior of steel–concrete composite-to-prestressed concrete bridge joints through ANSYS finite element analysis under negative bending moments.
- Performed parametric analyses of bearing-plate thickness, connector stiffness, and joint length, identifying the first row of PBL connectors as the critical shear-transfer region and establishing design recommendations for key structural parameters.

Design and Construction of FRP Pedestrian Bridge Using Recycled Wind Turbine Blades Oct 2022 – Aug 2024

Tongji University "Excellence Cup" / "Challenge Cup" Selection Competition, Team Leader

Supervisors: Prof. Qingtian Su & Prof. Xu Jiang (Civil Engineering, Tongji University) Oct. 2022 – Present

- Investigated the reuse of decommissioned wind turbine blades as load-bearing components for pedestrian bridges, assessing their mechanical properties and structural feasibility for bridge applications.
- Developed two pedestrian-bridge design schemes using recycled wind turbine blades and evaluated their structural performance under different loading conditions through 3D modeling and ABAQUS finite-element analysis.
- Won 3rd Prize in Tongji University's 8th "Excellence Cup" / 18th "Challenge Cup" Selection Competition and Silver Prize in the 9th China International "Internet+" Innovation and Entrepreneurship Competition for College Students.